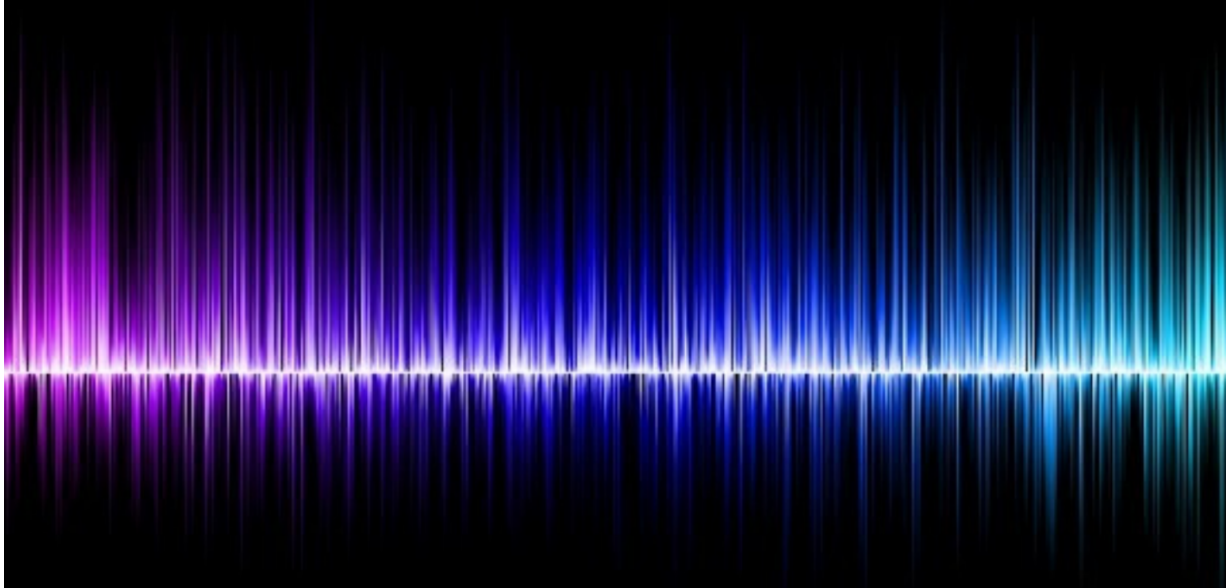


Sound Healing Certification

Science



LIFECHANGING
— ENERGY —

The brain is a complex organ. There are millions of neurons (brain cells) forming trillions of connections, in a symphony like no other. This symphony is what we call consciousness, and it is constantly changing as we respond to our environment.

The brain is made up of billions of neurons that communicate with each other through electrical impulses. These impulses are called brainwaves. Brainwaves are grouped into different categories depending on the frequency of these impulses. The slower the frequency, the deeper the sleep stage.

The brain is the command center of the body, directing all of its functions including movement, thinking, breathing, heart rate and the senses. These actions are controlled by electrical impulses that travel throughout the brain, neurons and body.

Our brains produce different types of electrical patterns known as brainwaves, which are measured in hertz (Hz), or cycles per second. Brainwave frequency determines what state we're in, from deep sleep to vigorous activity.

An electroencephalogram (EEG) measures brainwaves. When electrodes are placed on a person's head, they can detect these electrical impulses and send them to an amplifier. The amplifier then sends them to a computer where they can be viewed as waveforms on a graph.

Sound therapy or sound-based relaxation techniques are popular for helping people reduce stress, anxiety, and to fall asleep. Over time, these therapies can also help you change your brainwaves—the patterns of electrical activity. This is why sound therapy has been used for years to help people improve their mental health, overcome insomnia and generally improve their quality of life.

Brainwaves are named after their frequency.

Beta Brainwaves 14-40Hz — Awake Conscious & Reasoning

Beta brain waves are associated with normal waking state and a good sense of alertness, logic and critical reasoning.

Beta brain waves are important for effective functioning throughout the day, they can lead to stress, anxiety and overthinking.

Alpha 7.5-14Hz — Deep Relaxation Brainwaves

Alpha brain waves are present in deep relaxation, lovely daydream or during light meditation. Typically when the eyes are closed,

It is an optimal time to program the mind for success, flow up with affirmations and creative visualization.

Alpha brain state allow access to greater imagination, memory, learning and concentration.

Access your subconscious mind, base of your conscious awareness.

Alpha state get you in touch with your intuition, it becomes clearer and more profound the closer you get to 7.5Hz frequency.

Theta 4-7.5Hz — Deeper Meditation, Approaching Sleep

Theta brain waves are present during deep meditation and light sleep, including the all-important REM dream state. It is the realm of your subconsciousness and only experienced momentarily as you drift off to sleep from Alpha and wake from deep sleep (from Delta).

It is said that a sense of deep spiritual connection and unity with the universe can be experienced at Theta.

Your mind's most deep-seated programs are at Theta and it is where you experience vivid visualizations, great inspiration, profound creativity and exceptional insight. Unlike your other brain waves, the elusive voice of Theta is a silent voice.

It is at the Alpha-Theta border, from 7Hz to 8Hz, where the optimal range for visualization, mind programming and using the creative power of your mind begins. It's the mental state which you consciously create your reality. At this

frequency, you are conscious of your surroundings however your body is in deep relaxation.

Delta 0.5-4Hz — Deep Sleep and Restful Slumber

The Delta frequency is the slowest of the frequencies and is experienced in deep, dreamless sleep and in very deep, transcendental meditation where awareness is fully detached.

Delta is the realm of your unconscious mind, and the gateway to the universal mind and the collective unconscious, where information received is otherwise unavailable at the conscious level.

Among many things, deep sleep is important for the healing process — as it's linked with deep healing and regeneration. Hence, not having enough deep sleep is detrimental to your health in more ways than one.

Gamma 40Hz and Higher – Insight and Information Processing

This range is the most recently discovered and is the fastest frequency at above 40Hz. While little is known about this state of mind, initial research shows Gamma waves are associated with bursts of insight and high-level information processing.

**** Taken from <https://www.gjcae.org/contact-us/20-music/162-types-of-brainwaves-and-meditative-states>*

When your body perceives a particular type of sound it will vibrate at the same frequency as the sound. Your body's internal vibrations change in response to the frequencies that you hear from the outside world. That means the sounds that you hear impact what is happening inside your body.

Entrainment

The science behind sound therapy looks at frequencies. Every living thing has its own unique frequency, from an organ in your body to your own thoughts and feelings. When something goes wrong it means that the frequency has become disrupted and sound therapy can rebalance those frequencies by using different tones (musical notes) or by singing certain songs. The most basic example of this is using a tuning fork. If you place a tuning fork on one side of a glass then tap it,

you will find that water in the glass starts to vibrate at the same frequency as the tuning fork.

The science of entrainment is the study of the synchronization of oscillating systems. In other words, how one vibrating system can fall into step with another. Think of it as a group of people getting together and dancing in unison to the same tune. This phenomenon was first described in 1665 by Dutch scientist Christian Huygens, and he called it 'entrainment'.

Christian Huygens was the inventor of the pendulum clock and discovered that multiple clocks hanging from a common beam will tend to swing in unison after a short time.

Entrainment can apply to your heartbeat, your brainwaves or even your cells — when they start vibrating in harmony with some external rhythm, they synchronize their behavior and attune themselves to the stronger, more powerful vibration.

It's a naturally occurring phenomenon all around us, including inside your body, where some processes work in sync with others all the time.

Entrainment can occur between two or more independent rhythmic cycles (also known as oscillators). The rhythm of these oscillations match up and essentially become synchronized or entrained. An example is a pair of metronomes that are set to different tempos. They will continue to oscillate at different speeds but eventually start to synchronize with each other as long as they are close enough together. The frequency at which they will sync depends on the difference in beat cycle frequencies between the two metronomes.

Entrainment is everywhere. The sun and the moon. Your heart beat and the sound of music.

The principle of entrainment can be used for all kinds of purposes. For example, it's the basis for training in NLP (neuro-linguistic programming), hypnosis and meditation. In NLP training, an experienced practitioner uses specific language patterns to "entrain" his subject's brainwaves into different states that are conducive to re-programming the mind.

Hypnotherapists use entrainment techniques to help people feel relaxed and enter trance states during which they're highly suggestible and open to change. And meditators use breathing techniques or mantras ("ohm" is a popular one) to get their minds centered on a single object and achieve inner peace.

Entrainment happens in sound therapy and healing when someone hears a frequency and their body matches it.

Binaural Beats

Sound Therapy is based on the principle that all matter, including the human body, is in a continuous state of vibration. These vibrations are measured in hertz (Hz), with sound being measured in cycles per second.

When a sound or note is played, it sends out its own unique vibrational frequency. This frequency resonates throughout the body and can alter brainwaves which affect our moods, stress levels and even physical sensations like pain.

The theory of 'resonance' suggests that if an object is vibrating at the same frequency as another object, the two objects will start to resonate together. For example, if you were to play two pianos tuned to slightly different notes, eventually they would become tuned to each other due to resonance.

According to an article published in The Guardian, brain scans have revealed that listening to certain types of music can cause our bodies to release dopamine, a hormone responsible for feelings of pleasure or satisfaction.

"Dopamine has multiple effects on cognition and behavior," said Daniel Levitin, a neuroscientist at McGill University who studies the neuroscience of music. "It helps us focus attention and filter out distractions; it improves memory; it enhances our mood."

Other studies show that listening to music can calm the amygdala, an area of the brain that controls emotions like fear and anxiety. This may be why music therapy is so effective in treating mental health disorders like PTSD.

Sound therapy and healing involves the use of specific tones or frequencies that have an effect on our brain activity. It's based on the idea that exposure to particular sounds can lead to positive changes in our mood or behaviour.

One of the most popular forms of sound therapy is Binaural beats. Binaural beats are created when one tone of a specific frequency is played into one ear, and a slightly different tone is played into the other ear at the same time. The brain then creates a third tone that is equal to the difference between the two original frequencies. For example, if you listened to a sound in your left ear that had a frequency of 132 Hz, and a sound in your right ear that had a frequency of 140 Hz, your brain would process the difference between the two (8 Hz), and come to a new frequency — the desired one to alter your brain waves.

Binaural beats are not static or stable. They are constantly shifting and changing based on differences in the tones being played into each ear.

The use of binaural beats in healing goes back to the late 1800s when a German scientist named Heinrich Wilhelm Dove discovered them. However, it was not until 1973 that Dr. Gerald Oster published an article on them titled "Auditory Beats in the Brain" in Scientific American that they became widely known.

Let's talk about another example so we're clear ... Binaural beats are an auditory illusion perceived when two different pure-tone sine waves, both with frequencies lower than 1500 Hz, with less than a 40 Hz difference between them, are presented to a listener dichotically (one through each ear). For example, if a 530 Hz pure tone is presented to someone's right ear, while a 520 Hz pure tone is presented to the person's left ear, the listener will perceive the auditory illusion of a third tone, in addition to the two pure-tones presented to each ear. The third sound is called a binaural beat, and in this example would have a perceived pitch correlating to a frequency of 10 Hz, that being the difference between the 530 Hz and 520 Hz pure tones presented to each ear.

Binaural beats are created by listening to two slightly different frequencies at the same time through headphones. If done correctly, your brain will follow along and produce its own version of those frequencies internally.

Each frequency produces its own effect on your body. Depending on what you want to use them for, you will choose one or more types of binaural beats.

If you want to get technical, binaural beats are an auditory brainstem responses which originates in the superior olivary nucleus of each hemisphere. They result from the interaction of two different auditory impulses, originating in opposite ears. The difference between the two tones, is experienced as the two wave forms mesh in and out of phase within the superior olivary nuclei. This binaural beat is not heard in the ordinary sense of the word (remember the human range of hearing is from 20-20,000 Hz). It is perceived as an auditory beat and theoretically can be used to entrain specific neural rhythms through the frequency-following response (FFR).

Please note that A BINAURAL BEAT IS NOT A SOUND

A binaural beat does not exist.

Binaural beats are a type of auditory illusion created when two different frequencies are played through two different speakers or headphones. The brain then perceives the two sounds as one, and produces a binaural beat.

The benefits of binaural beats are numerous, but the two main ones are inducing relaxation and improving sleep. The brain needs to be in a relaxed state to fall asleep. When you listen to binaural beats, it triggers your brain to produce alpha and theta waves, which can induce a meditative and relaxed state.

Improving sleep is another great benefit of binaural beats. People often have trouble falling asleep because their mind is busy with racing thoughts or anxiety about the next day. Listening to binaural beats can help your brain relax and reduce those feelings of anxiety.

The benefits don't stop there. Binaural beats have been shown to reduce stress levels and improve mental focus by reducing distraction levels when studying or

working on projects. If you find it hard to focus on work, try listening to binaural beats in the background while you work on tasks that require concentration.

Binaural beats can also be used as an effective tool for meditation since they help the mind settle into a relaxed state by blocking out distractions, thus allowing you to reach a deeper state of consciousness faster than traditional meditation practices like sitting quietly and observing your breath.

It can even be used with voices and affirmations for positivity and manifestation techniques.

More recently, scientists have begun to explore the potential benefits of binaural beats, investigating the effects that these sounds may have on health and wellness.